NAME	Sheheryar Shakeel
ROLL NO	26K-3098

1. Task 1 (ATM Management System Using Functions): Write a C program that simulates a simple ATM system using five user-defined functions. Create welcome(), which takes no arguments and returns nothing, to display a welcome message. Create displayAmount(), which accepts a withdrawal amount, returns nothing, and displays the amount. Create getWithdrawalLimit(), which takes no arguments and returns the predefined maximum withdrawal limit of Rs. 50,000. Create calculateBalance(), which accepts the current account balance and the withdrawal amount and returns the remaining balance. Create checkWithdrawal(), which accepts the current balance, withdrawal amount, and maximum limit, and returns 1 if the withdrawal is valid or 0 otherwise. A withdrawal is valid only if the amount is greater than zero, does not exceed the account balance, and does not exceed the maximum withdrawal limit. In main(), call the functions as required, display the relevant information, and update the balance only if the withdrawal is valid.

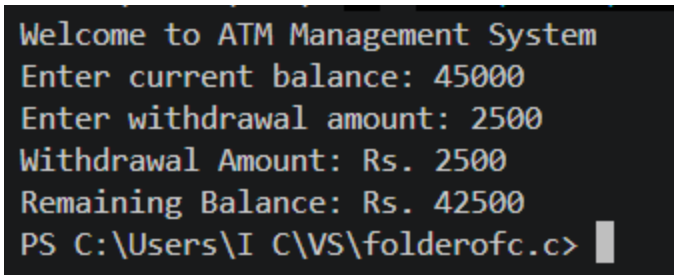
functions.c | func2.c

folderofc.c > functions.c

```c
#include <stdio.h>
void welcome(){
    printf("Welcome to ATM Management System\n");
}
void displayAmount(int amount){
    printf("Withdrawal Amount: Rs. %d\n", amount);
}
int getWithdrawalLimit(){
    return 50000;
}
int calculateBalance(int balance, int amount){
    return balance - amount;
}
int checkWithdrawal(int balance, int amount, int limit){
    if (amount > 0 && amount <= balance && amount <= limit){
        return 1;
    }
    else{
        return 0;
    }
}
int main(){
    int balance;
    int amount;
    int limit;
    int newBalance;
    welcome();
    printf("Enter current balance: ");
    scanf("%d", &balance);
    printf("Enter withdrawal amount: ");
    scanf("%d", &amount);

    limit = getWithdrawalLimit();

    if (checkWithdrawal(balance, amount, limit) == 1){
        displayAmount(amount);
        newBalance = calculateBalance(balance, amount);
        printf("Remaining Balance: Rs. %d\n", newBalance);
        balance = newBalance;
    }
    else{
        printf("Invalid Withdrawal!\n");
        printf("Maximum Withdrawal Limit: Rs. %d\n", limit);
    }
}
```



2. Task 2 (Shopping Management System Using Functions): Write a C program that simulates a shopping management system using user-defined functions. The program should display a welcome message, show a menu of available products with their prices, allow the user to select products and enter quantities, calculate the total price of each product, calculate the subtotal of all selected products, apply a discount if the subtotal reaches a specified amount, calculate the final bill, and display a shopping receipt. Create separate functions to display the welcome message, show the product menu, get the product price, calculate the price of an item, calculate the subtotal, calculate the discount, calculate the final bill, display the receipt, and ask the user whether they want to continue shopping. Use functions with different combinations of arguments and return types, including functions that take no arguments and return no value, functions that take arguments and return no value, functions that take no arguments and return a value, and functions that take arguments and return a value. Use appropriate loops and conditional statements to manage product selection, quantities, discounts, and repeated shopping. Display the selected products, quantities, individual prices, subtotal, discount, and final payable amount.

functions.c | func2.c

folderofc.c > func2.c

```c
#include <stdio.h>

void Welcome(){
    printf("Welcome Sir!\n");
    printf("Here is the menu!\n\n");
}
void ProductMenu(){
    printf("1. Pizza      500\n");
    printf("2. Burger     500\n");
    printf("3. Sandwich   450\n");
    printf("4. Roll       220\n");
}
int GetProductPrice(){
    int choice;
    printf("Enter product number: ");
    scanf("%d", &choice);
    if(choice == 1)
        return 500;
    else if(choice == 2)
        return 500;
    else if(choice == 3)
        return 450;
    else if(choice == 4)
        return 220;
    else
        return 0;
}
int CalculateItemPrice(int price, int quantity){
    return price * quantity;
}
int CalculateSubtotal(int subtotal, int itemPrice){
    return subtotal + itemPrice;
}
float CalculateDiscount(int subtotal){
    if(subtotal >= 1000)
        return subtotal * 0.10;
    else
        return 0;
}
float FinalBill(int subtotal, float discount){
    return subtotal - discount;
}
void Receipt(int subtotal, float discount, float finalBill){
    printf("\n--------------RECEIPT--------------\n");
    printf("Subtotal       : Rs. %d\n", subtotal);
```

```c
    printf("Subtotal       : Rs. %d\n", subtotal);
    printf("Discount       : Rs. %.2f\n", discount);
    printf("Final Payable  : Rs. %.2f\n", finalBill);
    printf("-----------------------------------\n");
}
int ContinueShopping(){
    int choice;

    printf("\nDo you want to continue shopping?\n");
    printf("1. Yes\n");
    printf("0. No\n");
    printf("Enter choice: ");
    scanf("%d", &choice);

    return choice;
}
int main(){
    int price, quantity;
    int itemPrice;
    int subtotal = 0;
    float discount;
    float finalBill;
    Welcome();
    do{
        ProductMenu();
        price = GetProductPrice();
        if(price == 0){
            printf("Invalid product!\n");
            continue;
        }
        printf("Enter quantity: ");
        scanf("%d", &quantity);
        itemPrice = CalculateItemPrice(price, quantity);
        printf("Price of selected item: Rs. %d\n", itemPrice);
        subtotal = CalculateSubtotal(subtotal, itemPrice);
    }
    while(ContinueShopping() == 1);
    discount = CalculateDiscount(subtotal);
    finalBill = FinalBill(subtotal, discount);
    Receipt(subtotal, discount, finalBill);
}
```

```
Welcome Sir!
Here is the menu!

1. Pizza      500
2. Burger     500
3. Sandwich   450
4. Roll       220
Enter product number: 2
Enter quantity: 2
Price of selected item: Rs. 1000

Do you want to continue shopping?
1. Yes
0. No
Enter choice: 0

--------------RECEIPT--------------
Subtotal       : Rs. 1000
Discount       : Rs. 100.00
Final Payable  : Rs. 900.00
-----------------------------------
PS C:\Users\I C\VS\folderofc.c>
```